

The polymorphic menagerie

Class hierarchies, relationally caged

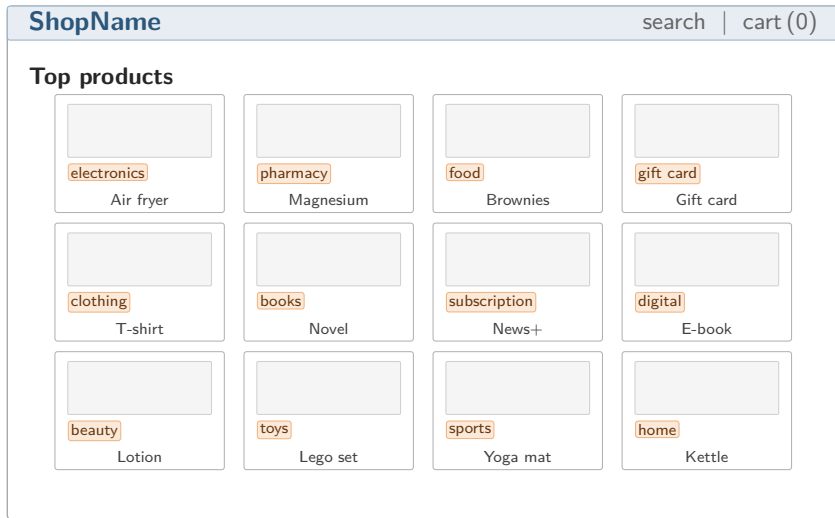
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pgEdge

2026



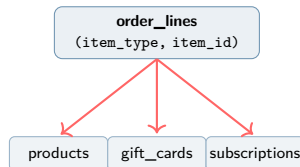
A page you've already seen



The menagerie's three habitats.

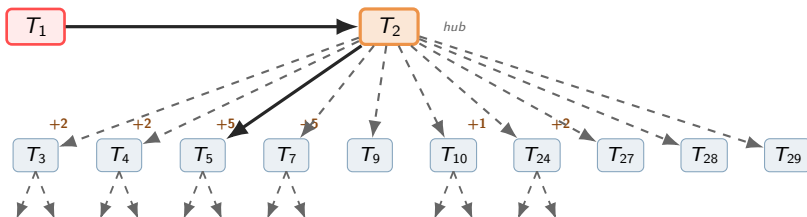
- **Catalogs** — e-commerce front pages, 1C Trade, ERP item registers
- **Activity / audit** — Salesforce WhoId/WhatId, Jira, audit logs
- **Documents / workflow** — SAP, Hibernate @Inheritance(JOINED), BPM tasks

Generated reflexively by ORMs (Rails, Django, Hibernate) and deliberately by platforms (Salesforce, 1C). Same query shape, same bottleneck.



No FK spans multiple targets
⇒ encoding workaround

What 1C actually generates



Legend:

- $\longrightarrow$ INNER JOIN (2 of them)
- $--\rightarrow$ LEFT OUTER JOIN (26 of them)

305 lines · 29 aliases · 28 joins · 0 EXISTS

($+N$ = grandchildren hidden; actual table types repeat — e.g. `_Reference189` appears 5 times)

The Query

```
SELECT
  ol.id,
  COALESCE(p.name, g.name, s.name) AS item_name
FROM order_lines ol
  LEFT JOIN products p
    ON ol.item_type = 'product' AND ol.item_id = p.id
  LEFT JOIN gift_cards g
    ON ol.item_type = 'gift_card' AND ol.item_id = g.id
  LEFT JOIN subscriptions s
    ON ol.item_type = 'subscription' AND ol.item_id = s.id
WHERE EXISTS (
  SELECT 1 FROM orders o
  WHERE o.id = ol.order_id AND o.placed_at >= 2024/01/01)
ORDER BY ol.popularity
LIMIT 100
```

*12 categories on the front page $\Rightarrow$ 12 candidate target tables in the schema;
we trace 3 here.*

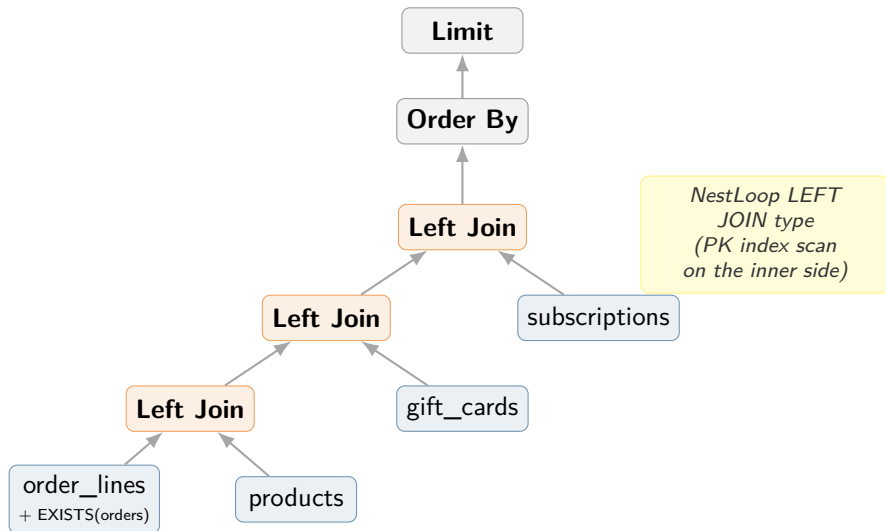
Why it blows up: the per-row mismatch

The discriminator picks one target per row.
The planner picks one probe order for *all* rows.

	products	gift_cards	subscriptions	
item_type='product'	✓	×	×	
item_type='gift_card'	×	✓	×	useful: 1/N
item_type='subscription'	×	×	✓	

wasted: $(N-1)/N$ per row

Query Plan Tree



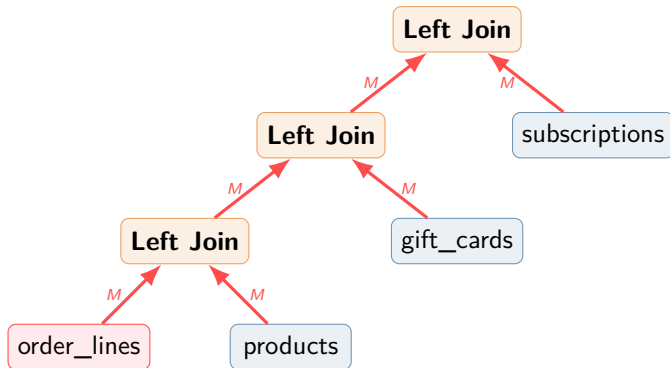
Main Execution Cost Amplifiers

- Base table (`order_lines`) growth
- Growth in the number of LEFT JOINs
- Ordering column of the base table not covered by an index
- EXISTS pull-up displaces the filter outside the optimisable block

Base Table Growth

*When M doubles, every probe count
doubles with it*

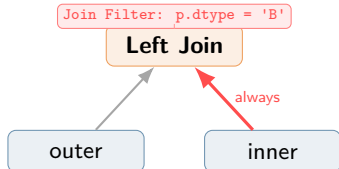
Amplifier: Base Table Growth



Every edge carries M rows. Doubling M doubles the number of probes on the inner side.

Concept: Result Filter

Current: probe, then filter



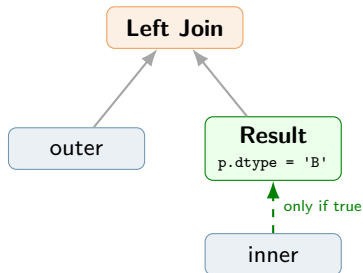
Inner scan runs **unconditionally**.

Filter rejects result *after* the probe.



Thread on pgsql-hackers

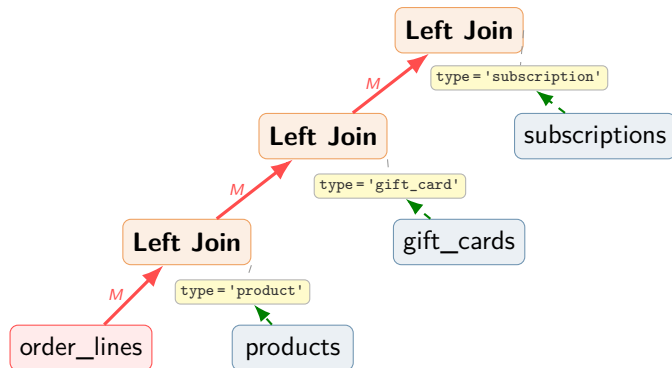
Proposed: Result filter atop RHS



Result node evaluates LHS-only clause as a **one-time filter**.

If false → returns $\emptyset$; inner scan *never launched*.

One-Sided Clause Guard



Guard checks discriminator *before* probing inner side. Each row probes only the matching table: $M_1 + M_2 + M_3 = M$.

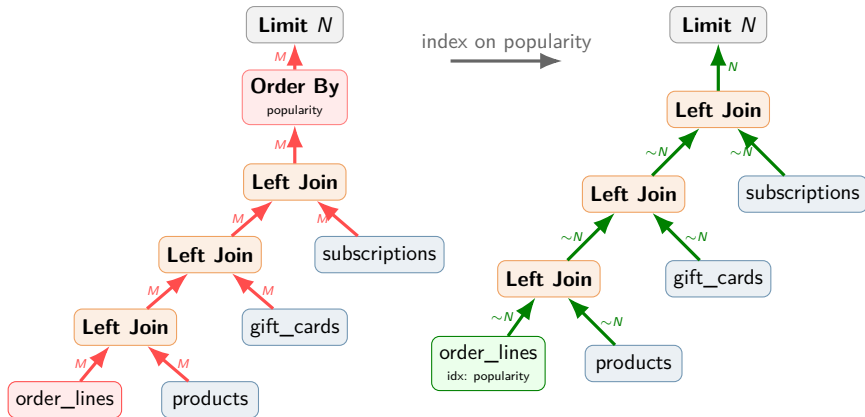
Benefits of the Result Filter Approach

- **Fewer page accesses.** Each base row probes only the *matching* inner table instead of all N . Total index/heap page reads drop from $M \times N$ to M .
- **Safety net for missing indexes.** When no index covers the inner side, the executor would fall back to a sequential scan *per outer row*. The guard lets it skip the scan entirely for non-matching rows.
- **Limiting price of a new LEFT JOIN.** Each new category in the model means one more LEFT JOIN at each query. With a guard clause, such new JOIN operator adds to the executor only strictly necessary actions without recursion into the inner subtree.

Aggregated Result

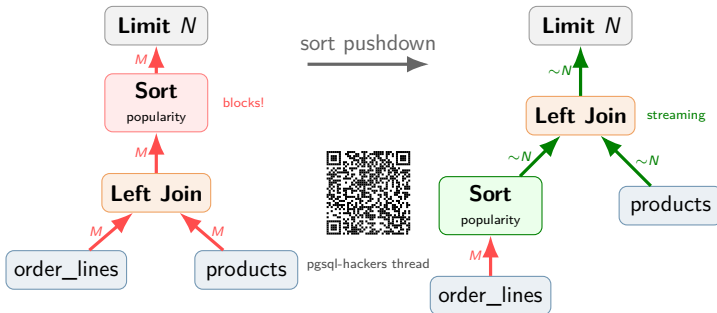
*ORDER BY + LIMIT,
and the blocking Sort that gets in their way*

ORDER BY + LIMIT: Effect of Proper Index

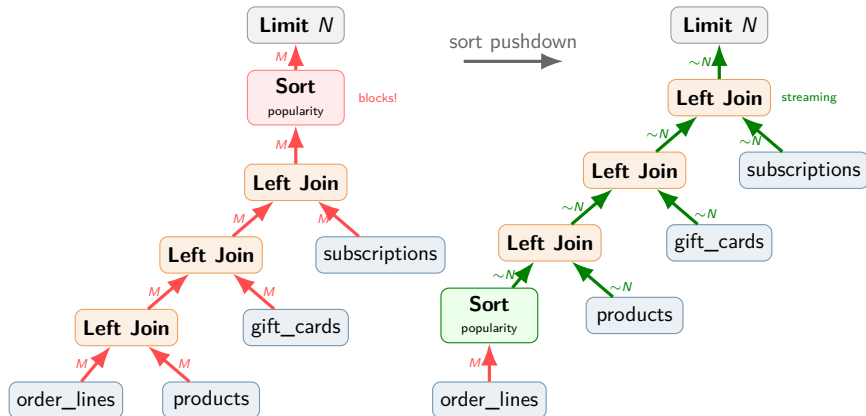


$N = LIMIT$. Index-ordered scan eliminates the blocking **Order By**. **Limit** stops the entire **LEFT JOIN** chain after N rows: $M \times 3 \rightarrow \sim N \times 3$ probes.

What if no proper index exists?



Sort Pushdown: Presorted Outer Path

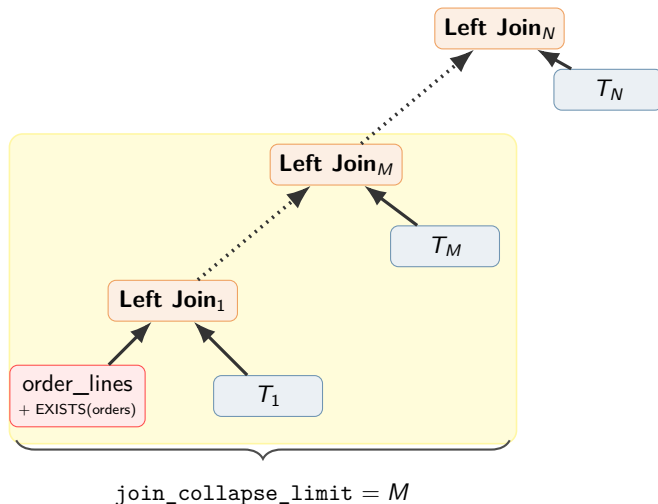


Push Sort below the joins: base rows sorted once ($M \log M$), but only $\sim N$ enter the join chain. Probes: $M \times 3 \rightarrow \sim N \times 3$.

The SubLink pull-up problem

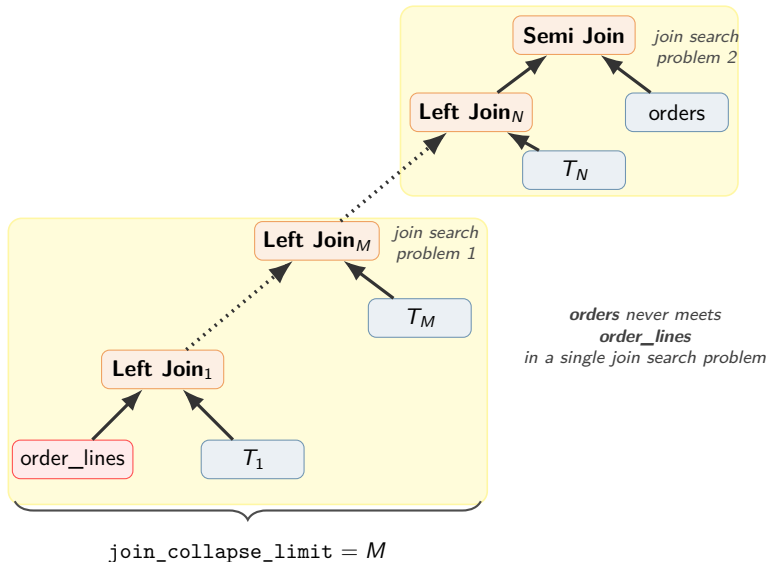
*EXISTS, the join_collapse_limit cliff,
and what happens between them*

An N -deep query tree



One problem — EXISTS lives on order_lines, inside the M-block.

An N -deep query tree after EXISTS pull-up



An N -deep query tree after SubLink relocation

